

Universidad de Salamanca
Degree in Physics



4th YEAR PHYSICS

**FINAL DEGREE PROJECT IN
PHYSICS**

**Monte Carlo simulation algorithms
for the Ising model**

Pablo López Serna
Supervisor: Antonio González Sánchez

Academic year 2025/2026

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Author's signature:

Signed: Pablo López Serna

Abstract

The Ising model is one of the paradigmatic systems in statistical mechanics, notable for the wide range of phenomena it displays despite its simple definition: a ferromagnetic–paramagnetic phase transition in 2D, the divergence of the correlation length at the critical temperature T_c , universality and critical exponents, and related phenomena such as hysteresis under applied fields.

In this work we study the two-dimensional Ising model using Monte Carlo simulations and compare three update algorithms: Metropolis, Glauber and Wolff. Each algorithm has advantages and limitations; the comparison aims to clarify which algorithm is preferable under different simulation regimes.

Simulations were performed on square lattices with side lengths $L \in \{16, 32, 64, 128\}$ and periodic boundary conditions, using $J = 1$ and temperature measured in units of J/k_B . For each size and temperature we collected sufficiently many samples after thermalization to estimate the observables of interest: the mean magnetization per spin $\langle|m|\rangle$, the heat capacity C , the magnetic susceptibility χ , the fourth-order Binder cumulant U_4 , and the second-moment correlation length ξ_L .

Critical exponents were estimated via finite-size scaling (FSS): if data are rescaled with the proper powers of L , curves for different sizes should collapse onto a single universal curve. We use visual collapse of χ and ξ_L to obtain approximate exponent values.

The Binder cumulant is scale-invariant and its crossings for different L provide an independent estimate of T_c without fitting exponents; we use this as a complementary method to reduce finite-size bias.

Algorithmic efficiency is assessed through the integrated autocorrelation time τ_{int} , computed with the Madras–Sokal windowing estimator (Madras y Sokal, 1988). Metropolis and Glauber show critical slowing down near T_c (large τ_{int}), while the Wolff cluster algorithm effectively eliminates this slowdown and yields very small autocorrelation times across the temperature range.

The numerical results agree with theoretical expectations (Onsager’s T_c and known critical exponents) and illustrate the trade-offs between algorithms in terms of accuracy and computational cost.

Keywords: Ising model, Monte Carlo, Metropolis, Glauber, Wolff, phase transition, finite-size scaling, critical exponents, critical temperature, critical slowing down.

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Chapter 1

Introduction

1.1 Motivation

Phase transitions are a central topic in statistical physics. A phase transition occurs when a system undergoes a change in its macroscopic properties as an external parameter, such as temperature, is varied. They are classified according to the behaviour of the free energy and its derivatives (Stanley, 1971): first-order transitions exhibit a discontinuity in the first derivative (latent heat and phase coexistence), while second-order (continuous) transitions show diverging second derivatives. This work focuses on a second-order transition, where the Ising model plays a paradigmatic role.

The Ising model was introduced by Wilhelm Lenz in 1920 and studied by Ernst Ising, who solved the one-dimensional case in 1925 (Ising, 1925). The absence of a phase transition in one dimension initially suggested that the model might be trivial in higher dimensions, but Lars Onsager’s exact solution for the two-dimensional model in 1944 demonstrated a finite-temperature second-order transition (Onsager, 1944; Peierls, 1936). The exact result for the square-lattice critical temperature is

$$T_c = \frac{2J}{k_B \ln(1 + \sqrt{2})} \approx 2,2692 \frac{J}{k_B}, \quad (1.1)$$

which we use as a reference for our numerical estimates. The availability of an exact benchmark makes the 2D Ising model an ideal testbed for validating simulation algorithms.

Beyond magnetism, the Ising model and its variants serve as minimal models for collective phenomena in diverse fields, from biology to social science. Given this broad relevance, this study develops and compares common Monte Carlo techniques to obtain numerical solutions, identifies the observables of interest, and contrasts the strengths and limitations of each algorithm.

1.2 Objectives

The main objective is a comprehensive comparison of three widely used Monte Carlo algorithms applied to the two-dimensional square-lattice Ising model without external field, for lattice sizes $L \in \{16, 32, 64, 128\}$. We implemented three Fortran programs that realise the Metropolis (Metropolis et al., 1953), Glauber (Glauber, 1963) and Wolff (Wolff, 1989) dynamics. To validate and compare the algorithms we measure the key

thermodynamic quantities: the mean magnetisation $\langle|m|\rangle$, the heat capacity per spin c , the magnetic susceptibility χ , the Binder cumulant U_4 , and the second-moment correlation length ξ_L , scanning both temperature and system size.

The temperature sweep resolves the behaviour around the critical region near T_c (eq. 1.1), while the size sweep enables finite-size scaling analysis to estimate critical exponents and extrapolate to the thermodynamic limit.

We also measure the integrated autocorrelation time. Although not a thermodynamic observable, τ_{int} quantifies dynamical efficiency and provides a useful comparison between algorithms, especially near criticality.

Document structure

The thesis is organised as follows. Chapter 2 presents the theoretical background required to understand the model, the phase transition and Monte Carlo methods, including the three update algorithms. Chapter 3 describes the simulated model and the numerical parameters. Results and the FSS analysis are presented in Chapter 4, and conclusions appear in Chapter 5.

Chapter 2

Theoretical background

In this chapter we present the theoretical background required to perform and interpret the simulations that follow. Before discussing Monte Carlo methods and specific update algorithms, we specify the system under study and the quantities used to characterise it. Starting from the physical origin of ferromagnetism and the ferromagnetic–paramagnetic phase transition, we introduce the Ising model and its Hamiltonian, define the relevant observables, and review the concepts (critical exponents, finite-size scaling, autocorrelation times) that underpin the analysis of the numerical results.

A ferromagnetic system is characterised by the alignment of microscopic magnetic moments (spins) in the same direction, producing a non-zero macroscopic magnetisation. Thermal fluctuations compete with this ordering: at low temperature spins tend to align, while at high temperature thermal disorder destroys the order. This competition gives rise to a critical temperature separating the ordered (ferromagnetic) and disordered (paramagnetic) phases. In the thermodynamic limit the transition is sharp, but in finite systems the singular behaviour is rounded and only approaches the ideal discontinuity as the system size increases.

We model the system in contact with a thermal reservoir at temperature T and describe it using the canonical ensemble. Below we derive the expressions used to compute the physical observables from simulation data, formally define the Ising Hamiltonian, state the approximations and boundary conditions, and introduce the algorithms employed in this work.

2.1 Ising model

We now give a complete definition of the Ising model used throughout this work. The system is a two-dimensional square lattice with a classical spin $s_i \in \{+1, -1\}$ placed on each lattice site. The general Hamiltonian reads

$$\mathcal{H} = -J \sum_{\langle i,j \rangle} s_i s_j - H \sum_i s_i,$$

where $s_i = \pm 1$ is the spin at site i , J is the coupling constant and H is a uniform external magnetic field. The first sum runs over nearest-neighbour pairs and the second over all sites.

For $J > 0$ the ground state is ferromagnetically ordered, while $J < 0$ favours antiferromagnetic order. In this work we study the ferromagnetic case without external field

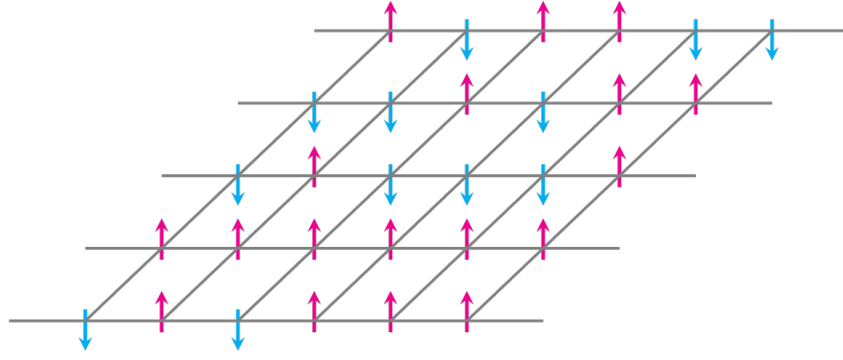


Figure 2.1: Example of a patch of the two-dimensional Ising lattice. Source.

($H = 0$), so the Hamiltonian simplifies to $\mathcal{H} = -J \sum_{\langle i,j \rangle} s_i s_j$ with nearest-neighbour interactions only.

The natural order parameter for the transition is the magnetisation per spin,

$$m = \frac{1}{N} \sum_i s_i,$$

with $N = L^2$ the total number of spins. In the thermodynamic limit $m = 0$ for $T > T_c$ and $m \neq 0$ for $T < T_c$. In finite systems global spin-inversion symmetry implies $\langle m \rangle = 0$ for all T , so in simulations one typically uses $\langle |m| \rangle$ as the estimator of the order parameter.

2.2 Mecánica estadística

In the canonical ensemble the equilibrium state of a system in contact with a heat bath at temperature T is specified by the partition function

$$Z = \sum_{\{\sigma\}} e^{-\beta \mathcal{H}(\sigma)},$$

with $\beta = 1/(k_B T)$ and the sum over all microstates $\{\sigma\}$. Thermodynamic quantities follow from Z ; for instance the free energy $F = -k_B T \ln Z$, and the average energy

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}.$$

The heat capacity per spin can be written in terms of energy fluctuations as

$$c = \frac{1}{N} \frac{\partial \langle E \rangle}{\partial T} = \frac{1}{N k_B T^2} (\langle E^2 \rangle - \langle E \rangle^2). \quad (2.1)$$

Similarly, the magnetic susceptibility per spin is related to magnetisation fluctuations,

$$\chi = \frac{1}{N k_B T} (\langle M^2 \rangle - \langle M \rangle^2),$$

with $M = \sum_i s_i$ the total magnetisation. These relations are used to compute c and χ from time-series produced by the simulations.

We also use the fourth-order Binder cumulant,

$$U_4 = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2},$$

which is dimensionless and takes a universal value at T_c depending only on the universality class and boundary conditions; this makes it a convenient tool to locate the critical temperature in finite systems (see Sec. 2.4).

The correlation length ξ_L measures the typical distance over which spin fluctuations are correlated and can be obtained from the spin–spin correlation function $g(r) = \langle s_0 s_r \rangle - \langle s_0 \rangle \langle s_r \rangle$, which decays approximately exponentially away from T_c , $g(r) \sim e^{-r/\xi_L}$. In finite periodic lattices it is practical to use the second-moment correlation length (Cooper et al., 1982; Newman y Barkema, 1999), built from the structure factor

$$\chi(\mathbf{k}) = \frac{1}{N} \left\langle \left| \sum_j s_j e^{i\mathbf{k} \cdot \mathbf{r}_j} \right|^2 \right\rangle, \quad (2.2)$$

evaluated at $\mathbf{k} = 0$ and at the smallest non-zero wavevector $k_{\min} = 2\pi/L$. The second-moment correlation length is then

$$\xi_L = \frac{1}{2 \sin(\pi/L)} \sqrt{\frac{\chi(0)}{\chi(k_{\min})} - 1}. \quad (2.3)$$

This definition is convenient because it can be computed directly from configurations sampled during the simulation without evaluating $g(r)$ for every distance.

Expected values appearing in the above formulas are phase averages,

$$\langle O \rangle = \frac{1}{Z} \sum_{\{\sigma\}} O(\sigma) e^{-\beta \mathcal{H}(\sigma)},$$

which are not computed directly by summation over the 2^N microstates but estimated via importance sampling. A Markov-chain Monte Carlo algorithm generates a sequence of configurations $\sigma_1, \dots, \sigma_M$ distributed according to the Boltzmann weight once equilibrium is reached. Provided the chain is ergodic and satisfies detailed balance (conditions enforced by the algorithms described in Sec. 3.6), phase averages are obtained as time averages over the sampled configurations:

$$\langle O \rangle \approx \frac{1}{M} \sum_{t=1}^M O(\sigma_t),$$

where σ_t denotes the full spin configuration at the t -th measurement after thermalisation and M is the number of measurements.

2.3 Phase transition, critical temperature and critical exponents

In the thermodynamic limit the 2D Ising model undergoes a phase transition exactly at T_c (eq. 1.1). Near this point thermodynamic quantities become non-analytic. In particular,

the correlation length diverges as

$$\xi(T) \sim |T - T_c|^{-\nu},$$

and other observables follow power laws with their characteristic exponents: the magnetisation behaves as $m \sim (T_c - T)^\beta$ for $T < T_c$, the susceptibility as $\chi \sim |T - T_c|^{-\gamma}$, and the heat capacity as $c \sim |T - T_c|^{-\alpha}$. For the 2D Ising model the heat-capacity divergence is only logarithmic, corresponding to $\alpha = 0$.

The exponents $\nu, \beta, \gamma, \alpha$ are called critical exponents and arise because at T_c the correlation length diverges and the system has no intrinsic length scale. Consequently the critical behaviour depends only on dimensionality and symmetry of the order parameter, not on microscopic details — the phenomenon of universality (Wilson, 1974). The 2D Ising model belongs to a well-known universality class with exact exponent values (Onsager and subsequent results (Yang, 1952)):

$$\nu = 1, \quad \gamma = \frac{7}{4}.$$

These values serve as references for comparing the numerical estimates from simulations.

2.4 Finite-size scaling

The divergences mentioned above occur only in the limit $L \rightarrow \infty$. In a finite system the correlation length cannot exceed the system size and saturates at $\mathcal{O}(L)$ near T_c , so thermodynamic quantities are smooth functions of T . Finite-size scaling (FSS) theory (Fisher y Barber, 1972) assumes that when $\xi \sim L$ the only relevant ratio is L/ξ , and any observable can be expressed as a scaling form depending on $(T - T_c)L^{1/\nu}$. For our observables

$$\begin{aligned} \langle |m| \rangle(L, T) &= L^{-\beta/\nu} f_m((T - T_c)L^{1/\nu}), \\ \chi(L, T) &= L^{\gamma/\nu} f_\chi((T - T_c)L^{1/\nu}), \end{aligned} \tag{2.4}$$

$$\xi_L(L, T) = L f_\xi((T - T_c)L^{1/\nu}), \tag{2.5}$$

where f_m, f_χ, f_ξ are universal scaling functions. Plotting appropriately rescaled observables (e.g. $\chi L^{-\gamma/\nu}$) against $(T - T_c)L^{1/\nu}$ for different L should collapse the curves onto a single universal function when the correct exponents and T_c are used, providing a direct test of universality.

Finite-size effects also shift pseudocritical temperatures: peak positions for χ or c occur at $T_c(L)$ satisfying

$$T_c(L) - T_c \sim L^{-1/\nu}.$$

The Binder cumulant $U_4 = 1 - \langle m^4 \rangle / (3\langle m^2 \rangle^2)$ (Binder, 1981) is dimensionless and scales as $U_4(L, T) = f_U((T - T_c)L^{1/\nu})$, so its curves for different L should cross at T_c , enabling an estimate of the critical temperature without fitting other exponents.

2.5 Autocorrelation time

We clarify the notion of "time" used here. Monte Carlo simulations do not reproduce real microscopic dynamics; configurations are generated by stochastic update rules. Therefore the "time" index is simply the sequence index measured in Monte Carlo steps (Sec. 3.2).

It is a discrete, dimensionless, algorithm-dependent unit. In what follows any temporal scale is measured in MCS.

The generated sequence $\sigma_1, \sigma_2, \dots, \sigma_M$ is temporally correlated since each configuration is obtained from the previous one by small updates. Correlations for an observable O are quantified by the autocorrelation function (Landau y Binder, 2014)

$$C_O(t) = \langle O(t_0)O(t_0 + t) \rangle - \langle O \rangle^2,$$

which typically decays exponentially $C_O(t) \sim e^{-t/\tau_O}$, where τ_O is the autocorrelation time associated to O . Larger τ_O means more MCS must be skipped between measurements to obtain effectively independent samples: the effective number of independent measurements is approximately $M/(2\tau_O)$ rather than M .

Near T_c , where ξ grows, autocorrelation times also increase (critical slowing down), described by the dynamic exponent z (Hohenberg y Halperin, 1977):

$$\tau \sim \xi^z \sim L^z.$$

Unlike static exponents, z depends on both universality class and the update algorithm. Local algorithms such as Metropolis or Glauber have relatively large z for the 2D Ising model ($z \approx 2.17$), while cluster algorithms (Swendsen y Wang, 1987) like Wolff reduce z substantially. This difference in $\tau(L)$ near criticality is a central aspect of our algorithmic comparison.

In practice we compute the normalized autocorrelation function

$$\rho_O(t) = \frac{C_O(t)}{C_O(0)},$$

and the integrated autocorrelation time

$$\tau_{\text{int},O} = \frac{1}{2} + \sum_{t=1}^{\infty} \rho_O(t), \tag{2.6}$$

which determines the true variance of the sample mean:

$$\text{Var}(\langle O \rangle) \approx \frac{2\tau_{\text{int},O}}{M} \text{Var}(O).$$

The sum in eq. (2.6) must be truncated in practice at some cutoff t^* where $\rho_O(t)$ has sufficiently decayed; choosing t^* requires care to avoid under- or overestimating $\tau_{\text{int},O}$ due to noise.

Chapter 3

Model and methods

3.1 Simulated model

We simulate the two-dimensional Ising model on a square lattice of $N = L \times L$ spins with periodic boundary conditions in both directions. Spins take values $\sigma_i = \pm 1$ and the Hamiltonian is given by eq. (??) with nearest-neighbour coupling $J = 1$ and no external field ($H = 0$). We set Boltzmann's constant $k_B = 1$, so it is convenient to work in reduced units

$$T^* = \frac{k_B T}{J},$$

which with our choice $J = k_B = 1$ equals T numerically; we drop the asterisk and denote the temperature simply by T . With this convention Onsager's exact critical temperature is $T_c \approx 2.2692$ Onsager (1944).

We study lattice sizes $L \in \{16, 32, 64, 128\}$ (i.e. $N = 256, 1024, 4096, 16384$ spins). This choice balances the need for a sufficiently wide size range (a factor of 8 between smallest and largest) for reliable finite-size scaling analysis with the computational cost, which scales as L^2 per Monte Carlo step; near T_c the number of steps required for independent samples also grows like L^z , making very large lattices impractical for local update algorithms.

3.2 Definition of a Monte Carlo step

For Metropolis and Glauber dynamics we define one Monte Carlo step (MCS) as $N = L^2$ single-spin flip attempts, i.e. on average one attempt per spin. Each attempted spin is chosen at random with replacement, so some spins may be proposed more than once and others not at all within the same MCS. This convention yields a time unit that is comparable across different system sizes.

For the Wolff cluster algorithm the cluster size varies between updates. We adopt the standard convention that one Wolff MCS consists of building as many clusters as needed until the total number of flipped spins in that MCS sums to N . On average this flips the same number of spins per MCS as the local-update algorithms, making time units comparable between all three methods. The number of clusters per MCS is inversely proportional to the mean cluster size: few clusters near T_c (large clusters) and many clusters far from T_c (small clusters).

3.2.1 Boundary conditions

Finite lattices have a boundary fraction of order $1/L$, which can distort results if spins at the edges have fewer neighbours than interior spins. To eliminate boundary effects we use periodic boundary conditions: right-edge spins are neighbours of left-edge spins and top-edge neighbours of bottom-edge spins, so the lattice is topologically a torus and every spin has the same number of neighbours. This restores translational homogeneity; the remaining deviations from the thermodynamic limit are the finite-size effects analysed with FSS (Sec. 2.4).

3.3 Initial configuration and thermalisation

For each combination of L , T and algorithm simulations start from the fully ordered configuration with all spins $+1$ (cold start). This choice avoids trapping in metastable regions when sweeping a wide temperature range. A hot start (random spins) converges faster in the paramagnetic phase but may be more prone to metastability; we use cold starts throughout to be conservative.

After the initial configuration we perform a number of thermalisation MCS without measurements so the system forgets the initial condition and reaches equilibrium at temperature T . The thermalisation length depends on temperature: in the coarse grid away from the critical region Metropolis and Glauber autocorrelation times are small and 6000 MCS suffice; within the fine window around T_c critical slowing down ($\tau \sim L^z$) requires longer thermalisation, so we use 10000 MCS there. Wolff's autocorrelation times are of order unity across the range, so 6000 MCS are more than enough for cluster updates.

3.4 Temperature grid

The temperature sweep is the same for all algorithms and sizes and spans $T = 0.15$ to $T = 5.0$. We use an adaptive resolution with a fine grid in a critical window and a coarser grid elsewhere.

The central critical window is $[T_{\text{lo}}, T_{\text{hi}}]$, asymmetric around T_c , with

$$T_{\text{lo}} = T_c - 1.0 \frac{T_c}{L}, \quad T_{\text{hi}} = T_c + 3.0 \frac{T_c}{L}.$$

The width scales as T_c/L because FSS predicts that finite-size structure (peak of χ , Binder crossing, inflection of ξ_L/L) concentrates in a region of relative width $\sim L^{-1/\nu} = L^{-1}$ around T_c (Newman y Barkema, 1999). The asymmetry (factor 1 below and 3 above T_c) reflects that in finite systems the susceptibility peak typically occurs slightly above T_c , so we allow extra margin on the high-temperature side.

Inside the fine window we sample with step $\Delta T = 0.1 T_c/L$. Since the width is $T_{\text{hi}} - T_{\text{lo}} = 4T_c/L$, the fine window is divided into 40 subintervals (41 points), independent of L . Outside the window we use a coarse uniform step $\Delta T = 0.1$ on $[0.15, T_{\text{lo}}]$ and $(T_{\text{hi}}, 5.0]$.

3.5 Simulation parameters

After thermalisation we perform $N_{\text{steps}} = 1.99 \times 10^6$ measurement MCS per temperature and size, which provides statistically significant samples for our analysis. We do not record a measurement every MCS to reduce storage and correlation: for Metropolis and

L	Fine window $[T_{\text{lo}}, T_{\text{hi}}]$	ΔT_{fine}	Coarse range	ΔT_{coarse}
16	[2,127, 2,695]	$1,418 \times 10^{-2}$	$[0,150, 2,127) \cup (2,695, 5,000]$	0,1
32	[2,198, 2,482]	$7,091 \times 10^{-3}$	$[0,150, 2,198) \cup (2,482, 5,000]$	0,1
64	[2,234, 2,376]	$3,546 \times 10^{-3}$	$[0,150, 2,234) \cup (2,376, 5,000]$	0,1
128	[2,251, 2,322]	$1,773 \times 10^{-3}$	$[0,150, 2,251) \cup (2,322, 5,000]$	0,1

Table 3.1: Temperature windows for each lattice size L , using $T_c = 2,2692$, $T_{\text{lo}} = T_c - T_c/L$ and $T_{\text{hi}} = T_c + 3T_c/L$. The fine window always contains 40 subintervals (41 points), independently of L .

Glauber we sample every 50 MCS, yielding 3.98×10^4 measurements per temperature and size, while for Wolff we sample every 5 MCS, yielding 3.98×10^5 measurements per temperature and size.

The sampling intervals were chosen based on typical autocorrelation times outside the critical region, where τ_{int} is a few MCS for Metropolis and Glauber. Sampling every 50 MCS usually ensures effectively independent measurements away from criticality, but near T_c autocorrelation times for the largest sizes exceed 50 MCS, so measurements remain correlated; this increases statistical errors but exploring extremely long runs is beyond the present work.

For Wolff a 5-MCS sampling interval suffices across the whole temperature range because τ_{int} is of order unity even near T_c .

3.6 Monte Carlo methods

Computing the partition function Z exactly requires summing over 2^N microstates, which is infeasible except for tiny lattices. Monte Carlo methods avoid this by generating a Markov chain of configurations whose stationary distribution is the Boltzmann weight $P(\sigma) = e^{-\beta\mathcal{H}(\sigma)}/Z$, so that typical configurations contributing most to thermal averages are sampled more often (importance sampling) (Metropolis et al., 1953; Sokal, 1997).

Convergence to the correct distribution is ensured if transition probabilities $W(\sigma \rightarrow \sigma')$ satisfy detailed balance

$$P(\sigma) W(\sigma \rightarrow \sigma') = P(\sigma') W(\sigma' \rightarrow \sigma),$$

and the chain is ergodic (any configuration reachable from any other in a finite number of steps). The three algorithms described below differ in how they propose candidate configurations σ' and in the transition probabilities, but all satisfy these requirements.

3.6.1 Metropolis algorithm

The Metropolis algorithm (Metropolis et al., 1953) proposes local single-spin flips. From a configuration σ choose a site i uniformly at random and compute the energy change ΔE that would result from flipping $s_i \rightarrow -s_i$. Accept the new configuration with probability

$$W(\sigma \rightarrow \sigma') = \min(1, e^{-\beta\Delta E}).$$

If $\Delta E \leq 0$ the flip is always accepted; if $\Delta E > 0$ it is accepted with probability $e^{-\beta\Delta E}$, implemented by comparing a uniform random number in $[0, 1]$ to $e^{-\beta\Delta E}$. Rejected pro-

posals leave the configuration unchanged but still count as an attempt. Detailed balance follows because $W(\sigma \rightarrow \sigma')/W(\sigma' \rightarrow \sigma) = e^{-\beta\Delta E} = P(\sigma')/P(\sigma)$.

On the square lattice ΔE can take only five values (depending on the number of aligned neighbours), so acceptance probabilities can be precomputed to reduce floating-point operations.

3.6.2 Glauber dynamics

Glauber dynamics (Glauber, 1963) is another single-spin update scheme with a different transition rule. Instead of proposing a flip and accepting or rejecting it, Glauber updates the spin i by sampling its new value directly from the conditional probabilities given neighbour orientations. If E_i^+ and E_i^- are the energies with $s_i = +1$ and $s_i = -1$, the probability to set $s_i = +1$ is

$$p_i^+ = \frac{e^{-\beta E_i^+}}{e^{-\beta E_i^+} + e^{-\beta E_i^-}},$$

and $1 - p_i^+$ for $s_i = -1$. This rule also satisfies detailed balance and yields the same equilibrium distribution as Metropolis, but the dynamics differ: in Glauber the new state is chosen independently of the previous one, while Metropolis depends on the acceptance criterion. Both are local-update algorithms and share similar qualitative critical dynamics.

3.6.3 Wolff cluster algorithm

The Wolff algorithm (Wolff, 1989) belongs to the family of cluster algorithms designed to reduce autocorrelation times near T_c . Instead of updating single spins, a connected cluster of like-oriented spins is built and flipped as a whole.

The procedure is: pick a random seed spin and examine its neighbours; add each neighbour with the same orientation to the cluster with probability

$$p_{\text{add}} = 1 - e^{-2\beta J}.$$

Newly added spins are examined recursively until no further neighbours are added. The entire cluster is then flipped. This cluster acceptance probability preserves detailed balance for the global cluster flip, and the equilibrium distribution matches that of the local algorithms.

The advantage of Wolff is that near T_c clusters grow large and each update flips a macroscopic fraction of the lattice, dramatically reducing autocorrelation times. The dynamic exponent z is therefore much smaller than for Metropolis or Glauber, which is why we include Wolff in the comparison: it enables efficient sampling close to criticality for the lattice sizes used here.

3.7 Measured observables

At every measurement step we record the following quantities from the instantaneous configuration:

- Total energy of the system, $E = \mathcal{H}$.
- Signed magnetisation per spin, $m = \frac{1}{N} \sum_i \sigma_i$.
- Powers m^2 and m^4 , used to compute χ and U_4 .

- Structure factor at the smallest wavevector, $\chi(k_{\min})$, needed for ξ_L .

After completing the measurements at a given temperature we compute averages of these quantities and store them. Post-processing generates the final observables and plots. The measured quantities are as follows.

The mean absolute magnetisation per spin is defined as $\langle|m|\rangle$, i.e. the average of the absolute value of the magnetisation per spin. In a finite system global spin-inversion symmetry implies $\langle m \rangle \approx 0$, so using $\langle|m|\rangle$ provides a non-zero estimator of the order parameter.

The heat capacity per spin is computed from energy fluctuations:

$$C = \frac{\beta^2}{N} (\langle E^2 \rangle - \langle E \rangle^2),$$

which is eq. (2.1). This estimator is efficient because $\langle E^2 \rangle$ and $\langle E \rangle$ are accumulated during the same measurement pass.

The magnetic susceptibility per spin follows from magnetisation fluctuations:

$$\chi = \frac{L^2}{T} (\langle m^2 \rangle - \langle|m|\rangle^2),$$

using $\beta = 1/T$ and $M = Nm$, so $\chi = \beta(\langle M^2 \rangle - \langle|M|\rangle^2)/N = L^2(\langle m^2 \rangle - \langle|m|\rangle^2)/T$. In the disordered phase $\langle|m|\rangle \approx 0$ and $\chi \approx L^2\langle m^2 \rangle/T$.

The Binder cumulant is computed as

$$U_4 = 1 - \frac{\langle m^4 \rangle}{3\langle m^2 \rangle^2}.$$

In the ordered phase ($T \rightarrow 0$) $|m| \rightarrow 1$ and $U_4 \rightarrow 2/3$, while in the disordered phase ($T \rightarrow \infty$) m is approximately Gaussian with $U_4 \rightarrow 0$. Intersections of $U_4(T)$ curves for different L (interpolated using splines) are used to estimate T_c .

The second-moment correlation length ξ_L is computed from eq. (2.3), requiring the structure factor at $\mathbf{k}_1 = (2\pi/L, 0)$. In the Fortran simulations we accumulated real and imaginary parts of

$$\sum_j \sigma_j e^{i\mathbf{k}_1 \cdot \mathbf{r}_j}$$

separately and stored the averaged squared modulus per spin to obtain $\chi(k_{\min})$.

The integrated autocorrelation time τ_{int} is estimated for the time series of $|m(t)|$, the observable most sensitive to correlations near T_c . We use the Madras–Sokal windowing estimator described in Sec. 2.5 (eq. (2.6)), with the window W chosen automatically as the first time where $\rho(t) < e^{-t/\tau_{\text{int}}}$ and τ_{int} computed iteratively. Note that τ_{int} is a dynamical quantity, not an equilibrium observable, which is why its analysis is presented separately in Chapter 4.

3.8 Finite-size scaling analysis

To estimate T_c we first use Binder cumulant crossings. Data for each L are interpolated with splines, pairwise crossings between consecutive sizes are located, and the crossing temperatures are plotted against $1/L^2$. A linear extrapolation in $1/L^2$ should intersect

the vertical axis at T_c , yielding the estimate reported below while attempting to reduce finite-size corrections.

For the data-collapse analysis we use Onsager's exact value $T_c = 2.2692$ as a reference and the exact static exponents for the 2D Ising universality class: $\nu = 1$, $\gamma/\nu = 7/4$ and $\beta/\nu = 1/8$. We then rescale susceptibility and correlation-length data for all sizes and adjust exponents visually until an approximate collapse onto a single universal curve is observed.

We work with the reduced temperature $t = (T - T_c)/T_c$ and with the scaling variable $x = t L^{1/\nu} = (T - T_c)L/T_c$ (with $\nu = 1$). We obtain the susceptibility collapse by plotting $\chi L^{-\gamma/\nu} = \chi L^{-7/4}$ versus x : according to eq. (2.4) curves for different L should collapse. The same procedure applies to ξ_L/L versus x , which by eq. (2.5) should also collapse onto a universal function. We inspect collapses visually because a quantitative assessment would require an explicit model for the scaling function $\tilde{\chi}(x)$, beyond the scope of this work. With this choice the fine temperature window $[T_c - T_c/L, T_c + 3T_c/L]$ maps to a fixed interval $x \in [-1, 3]$, providing the same range and point density for all sizes.

Chapter 4

Results and discussion

This chapter presents the numerical results of the simulations described above and compares them with theoretical expectations. For each observable we first show the raw data for each size and algorithm and then perform the appropriate analysis, either qualitative or via FSS. The chapter closes with the dynamical comparison using integrated autocorrelation times, where the most significant differences between local and cluster dynamics — and between Metropolis and Glauber — appear.

4.1 Heat capacity and magnetisation

Figures 4.1 and 4.2 display the heat capacity c and the mean absolute magnetisation $\langle|m|\rangle$ as functions of T for the three algorithms and four lattice sizes.

As expected, the magnetisation per spin decreases smoothly (not abruptly) from values close to unity to nearly zero in the paramagnetic phase. This crossover sharpens with increasing L , approaching the discontinuous jump of the thermodynamic limit. The three algorithms yield almost identical results for these observables, which gives confidence that the Boltzmann ensemble has been sampled correctly. For the heat capacity the peak broadens and shifts to higher temperatures as L decreases, consistent with theoretical expectations.

One might attempt to estimate T_c by following the shift of heat-capacity maxima, but because the divergence of c in the 2D Ising model is only logarithmic, the peak height grows very slowly, making such extrapolations unreliable. This is a well-known difficulty and motivates the use of the Binder cumulant and the susceptibility in the FSS analysis.

The mean absolute magnetisation matches theoretical predictions far from criticality. Close to T_c the transition is rounded over a finite temperature window in finite systems, but the trend toward a sharper change with increasing L is clearly visible.

4.2 Binder cumulant

The Binder cumulant U_4 is a direct observable to locate the critical temperature: curves for different L should cross at T_c . In practice finite-size effects shift the crossings, so we use the linear-extrapolation method described in the text to estimate the $L \rightarrow \infty$ limit. Figure 4.3 shows the Binder cumulant near T_c and the extrapolated crossing temperatures obtained with this method.

The universal value of the Binder cumulant at the crossing for the 2D Ising model with

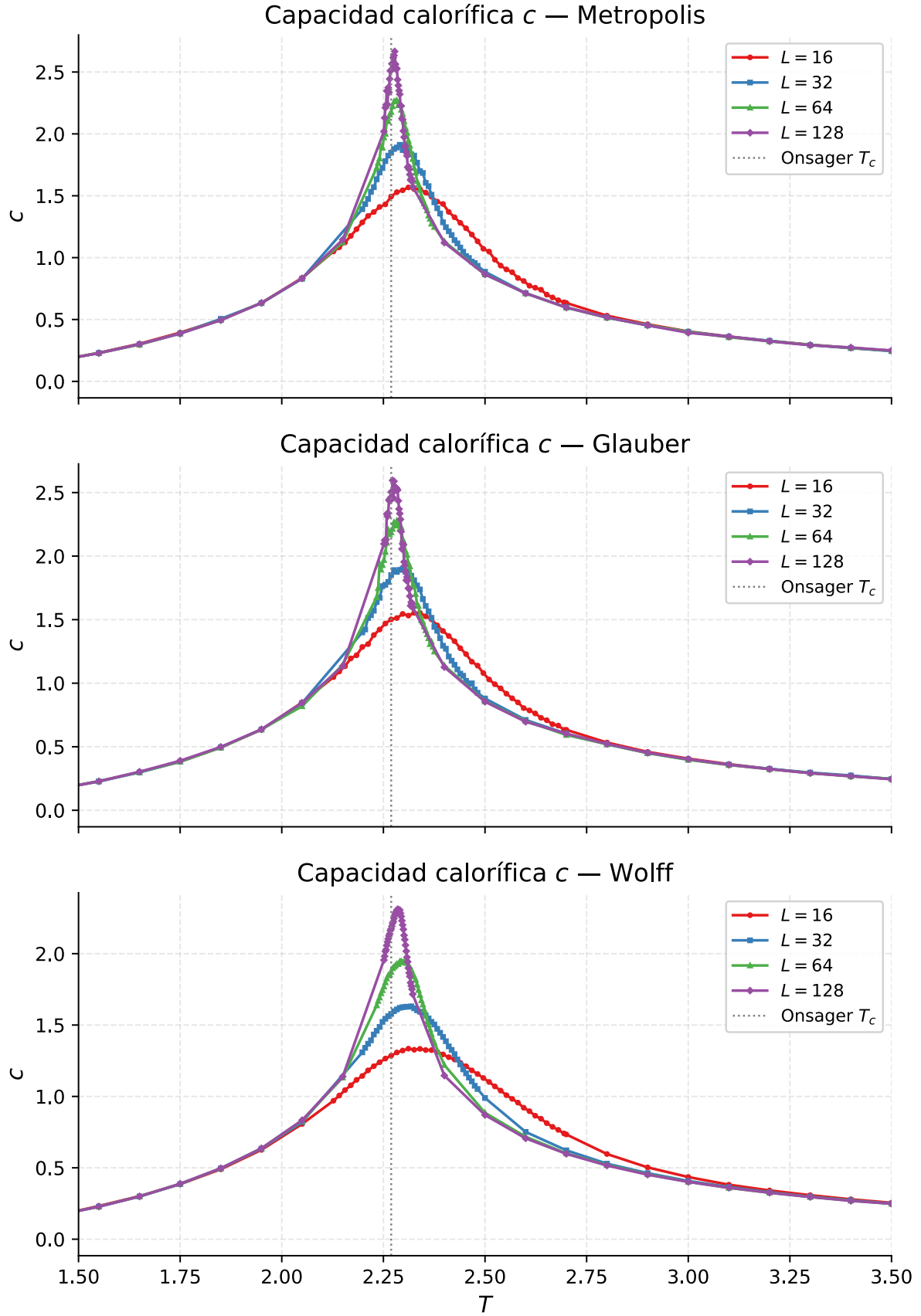


Figure 4.1: Heat capacity vs temperature for the three algorithms and $L \in \{16, 32, 64, 128\}$. The peak of c sharpens and moves toward T_c as L increases.

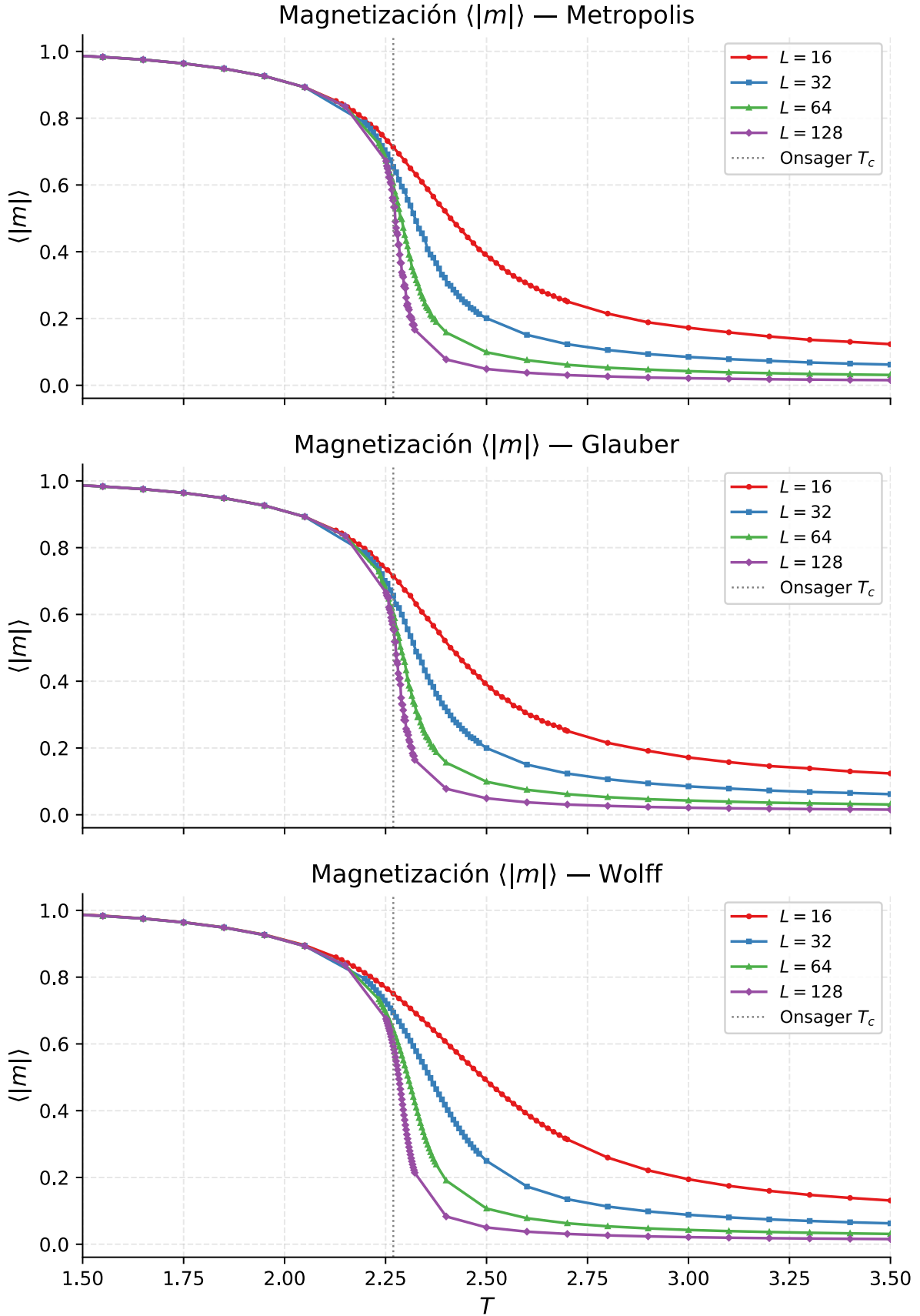


Figure 4.2: Mean magnetisation per spin as a function of temperature for the three algorithms and $L \in \{16, 32, 64, 128\}$.

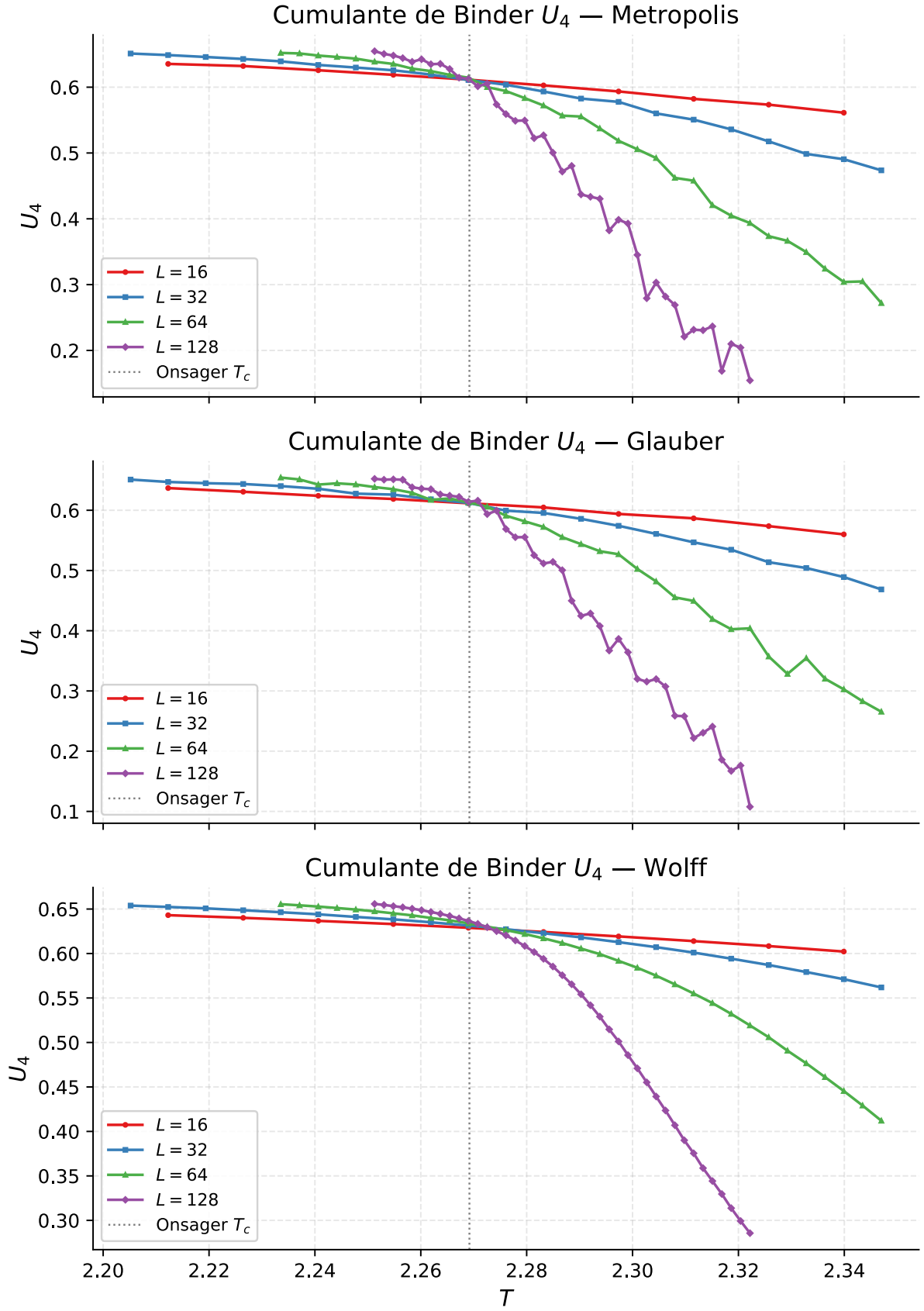


Figure 4.3: Binder cumulant for all algorithms and lattice sizes.

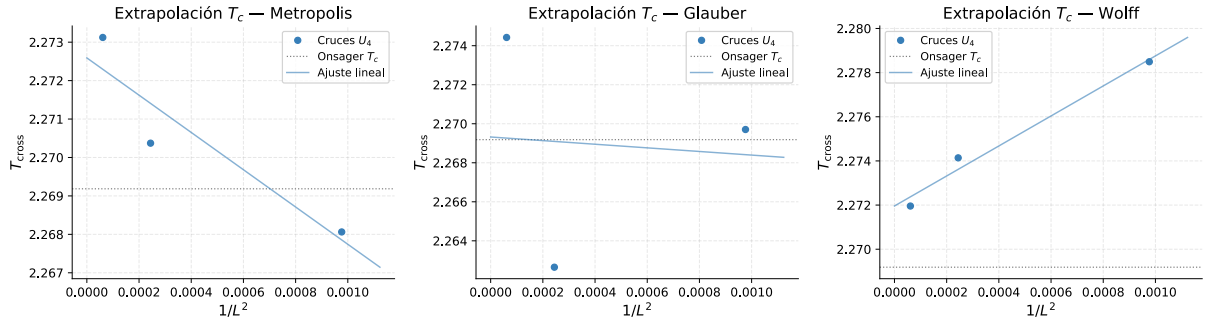


Figure 4.4: Extrapolation of U_4 crossing temperatures to the thermodynamic limit for each algorithm.

periodic boundary conditions is approximately $U_4^* \approx 0.61$ (Binder, 1981). This value is independent of L in the thermodynamic limit and provides an additional check: if simulation crossings occur near this value, the data are consistent with the expected universality class. In the figures the crossings indeed occur roughly at this value for all size pairs and algorithms, indicating the simulations and measurements are correct. Figure 4.4 shows the fitting lines for each algorithm together with the extrapolated T_c .

The finite-size correction to the crossing position scales as $L^{-1/\nu} = L^{-1}$ for the 2D Ising model, so doubling L should halve the systematic shift and the sequence $T_c(L, 2L)$ should converge smoothly to T_c . In Metropolis and Glauber the expected convergence is obscured by much larger statistical noise because their integrated autocorrelation times τ_{int} grow strongly near criticality, while Wolff does not suffer from this effect. As a result systematic improvements with increasing L are only evident for Wolff; Metropolis and Glauber data scatter more due to insufficient decorrelation.

4.3 Magnetic susceptibility

Figure 4.5 shows the susceptibility χ for the three algorithms and four sizes. The peak height clearly grows with L as expected from $\chi_{\text{max}} \sim L^{\gamma/\nu}$ with $\gamma/\nu = 7/4$, and the peak narrows and shifts slightly toward T_c , consistent with FSS predictions.

We can quantify the growth of the peak with L as an additional check. Extracting $\chi_{\text{max}}(L)$ for each size and plotting $\ln \chi_{\text{max}}$ vs $\ln L$ should yield a slope $\gamma/\nu = 7/4 \approx 1.75$. The linear fits shown provide an independent verification of the power-law $\chi_{\text{max}} \sim L^{\gamma/\nu}$.

4.4 Finite-size scaling of the susceptibility

Figure 4.7 shows the FSS collapse of the susceptibility using the experimentally obtained exponents for each algorithm.

The quality of the collapse is suboptimal in all cases. For Metropolis and Glauber the collapse may appear better but noisier for larger L , with their peaks roughly aligning near $x \approx 1$. The Wolff peak, however, is noticeably shifted to about $x \approx 1.5$. Since Wolff does not suffer critical slowing down, this discrepancy suggests that the runs for Metropolis and Glauber underestimated the number of steps needed to obtain effectively independent measurements, explaining the large differences between the local algorithms and Wolff.

This demonstrates that local-update algorithms fail to produce fully decorrelated samples near T_c even when the chosen thermalisation appears adequate. Section 4.6 discusses this

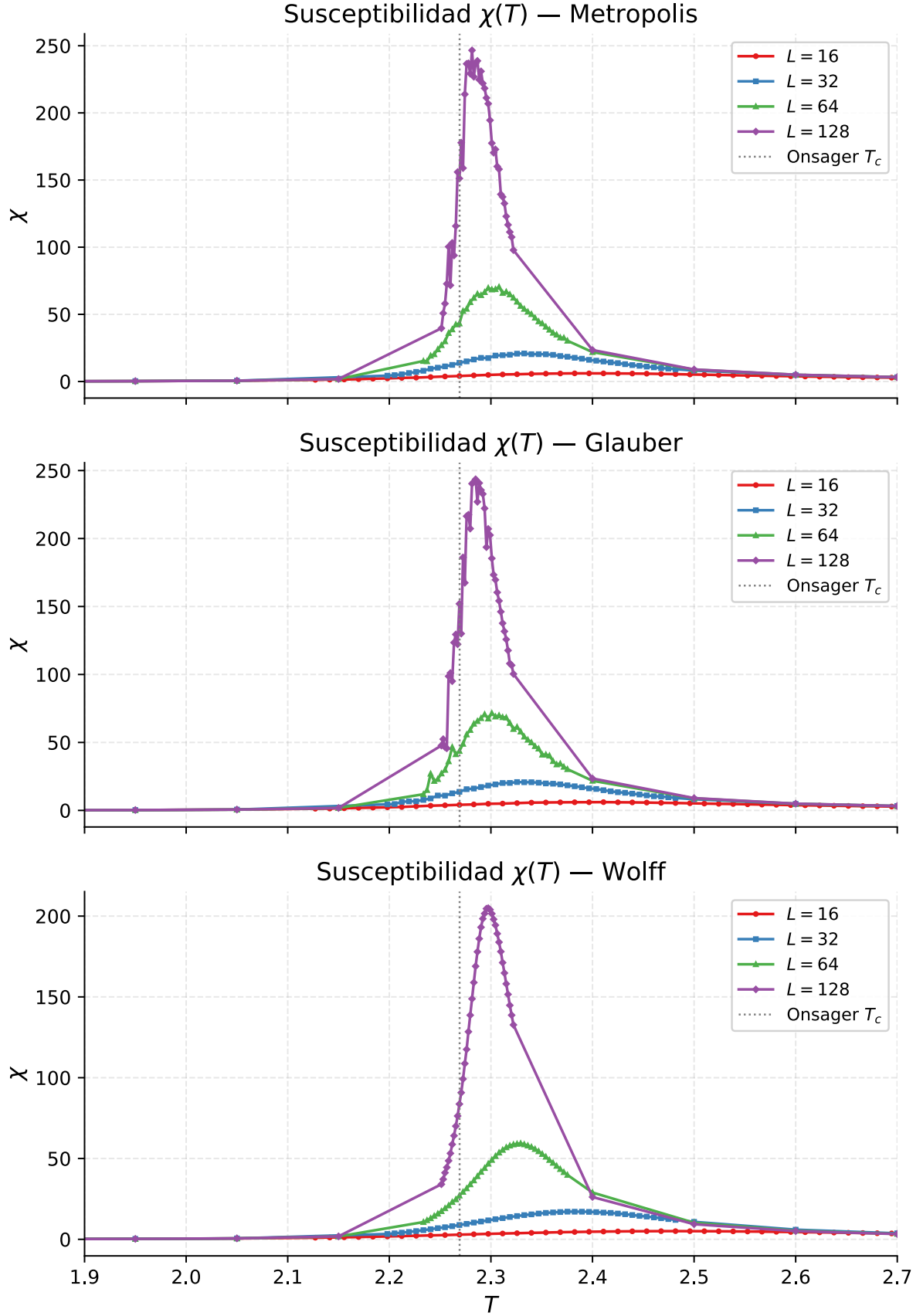


Figure 4.5: Magnetic susceptibility χ vs T for the three algorithms and $L \in \{16, 32, 64, 128\}$. The peak height increases following $L^{\gamma/\nu}$.

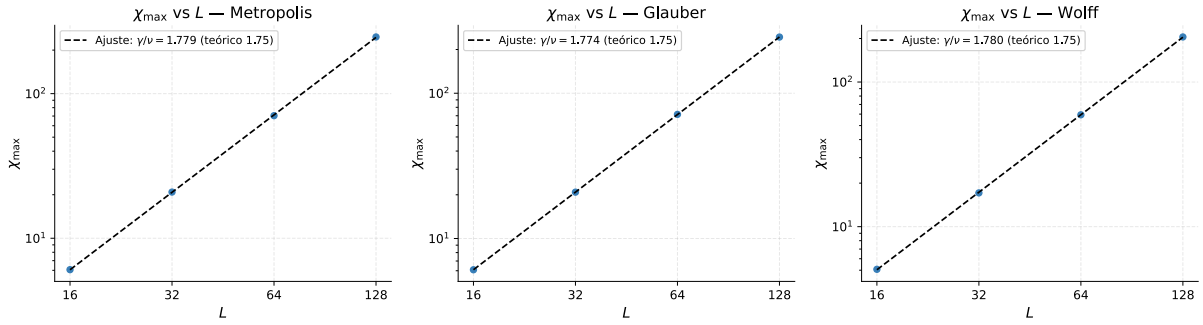


Figure 4.6: Linear fit of $\ln \chi_{\max}$ vs $\ln L$ for the algorithms and the fitted slope.

issue in more detail. Given the data quality, the extracted exponents are not highly precise for any algorithm.

4.5 Correlation length

Figure 4.8 displays ξ_L vs T for each algorithm. Values remain below $L/4$ for all sizes, as expected for the second-moment definition in the ordered phase; this does not indicate a problem since the ratio of peak heights follows the ratio of L .

The key quantity is the ratio ξ_L/L , shown in Fig. 4.9. The curves cross near T_c , similarly to the Binder cumulant but based on a different observable: the coincidence of crossing points indicates that the correlation length scales linearly with system size at criticality.

The fact that both U_4 and ξ_L/L cross at approximately the same temperature is a non-trivial internal consistency check: these observables are constructed in very different ways (moments of magnetisation vs structure factor), yet point to the same T_c . Agreement of the three algorithms on the crossing further supports that equilibrium was sampled correctly. A linear fit to the maxima positions yields the exponent $1/\nu$, shown in Fig. 4.10.

As expected, the fitted slope is close to one in all cases, indicating $\nu \approx 1$.

4.5.1 FSS collapse of the correlation length

Figure 4.11 presents the FSS collapse of ξ_L/L using the values of ν obtained from the collapse search for each algorithm.

The crossing visible in Fig. 4.9 becomes a collapse onto a common curve when plotted against $(T - T_c)L^{1/\nu}$, confirming $\nu = 1$ without additional fitting. Nevertheless, the collapse of ξ_L/L suffers from the same difficulties as the susceptibility: Metropolis and Glauber data become noisier for larger L , although a reasonable rescaling is still possible. For Wolff, despite cleaner curves, achieving a satisfactory collapse for any single ν value is challenging due to the observed systematic shifts.

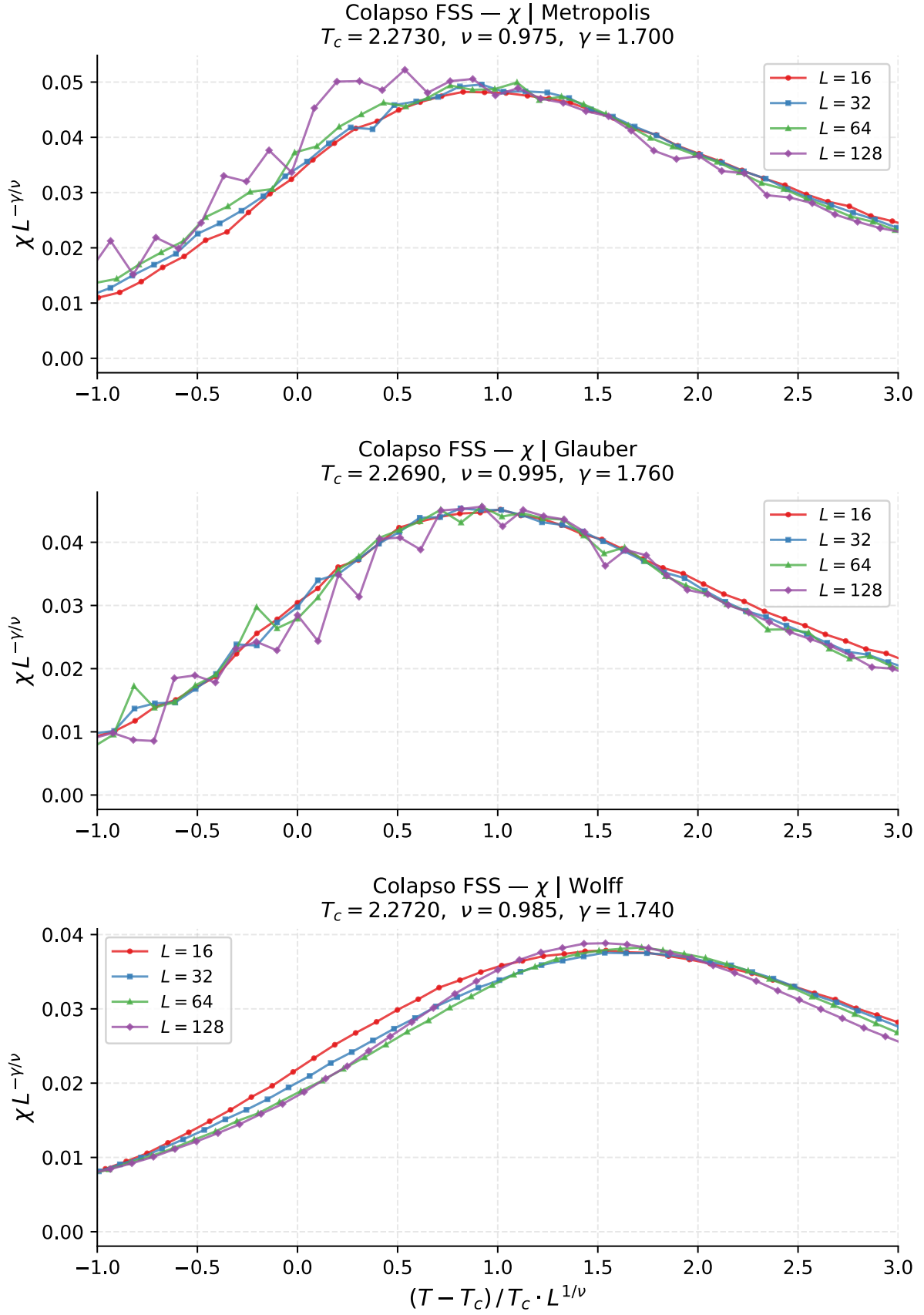


Figure 4.7: FSS collapse of χ using experimental exponents for each algorithm.

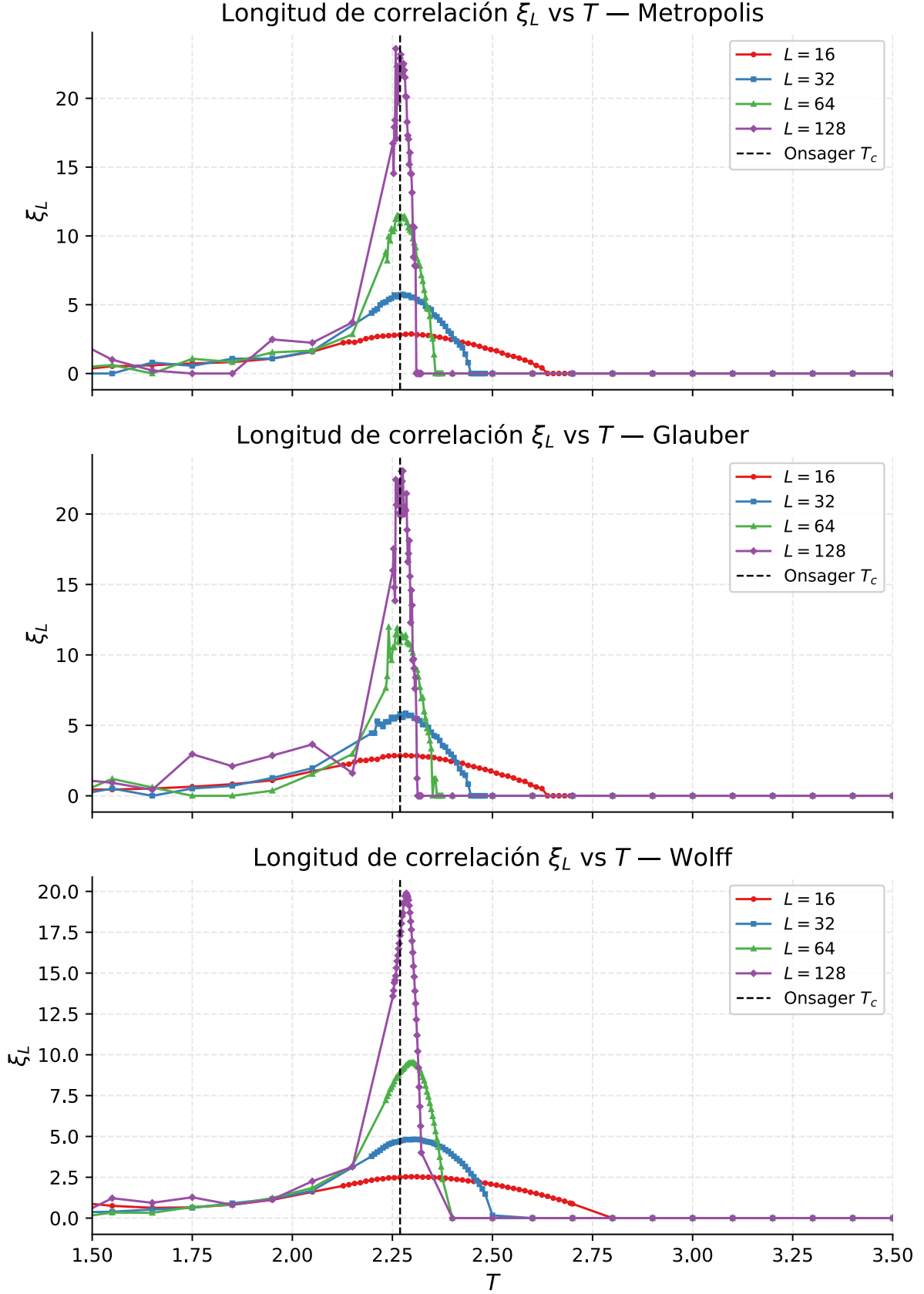


Figure 4.8: Second-moment correlation length ξ_L vs T for each algorithm and $L \in \{16, 32, 64, 128\}$. Peaks grow with L and shift towards T_c .

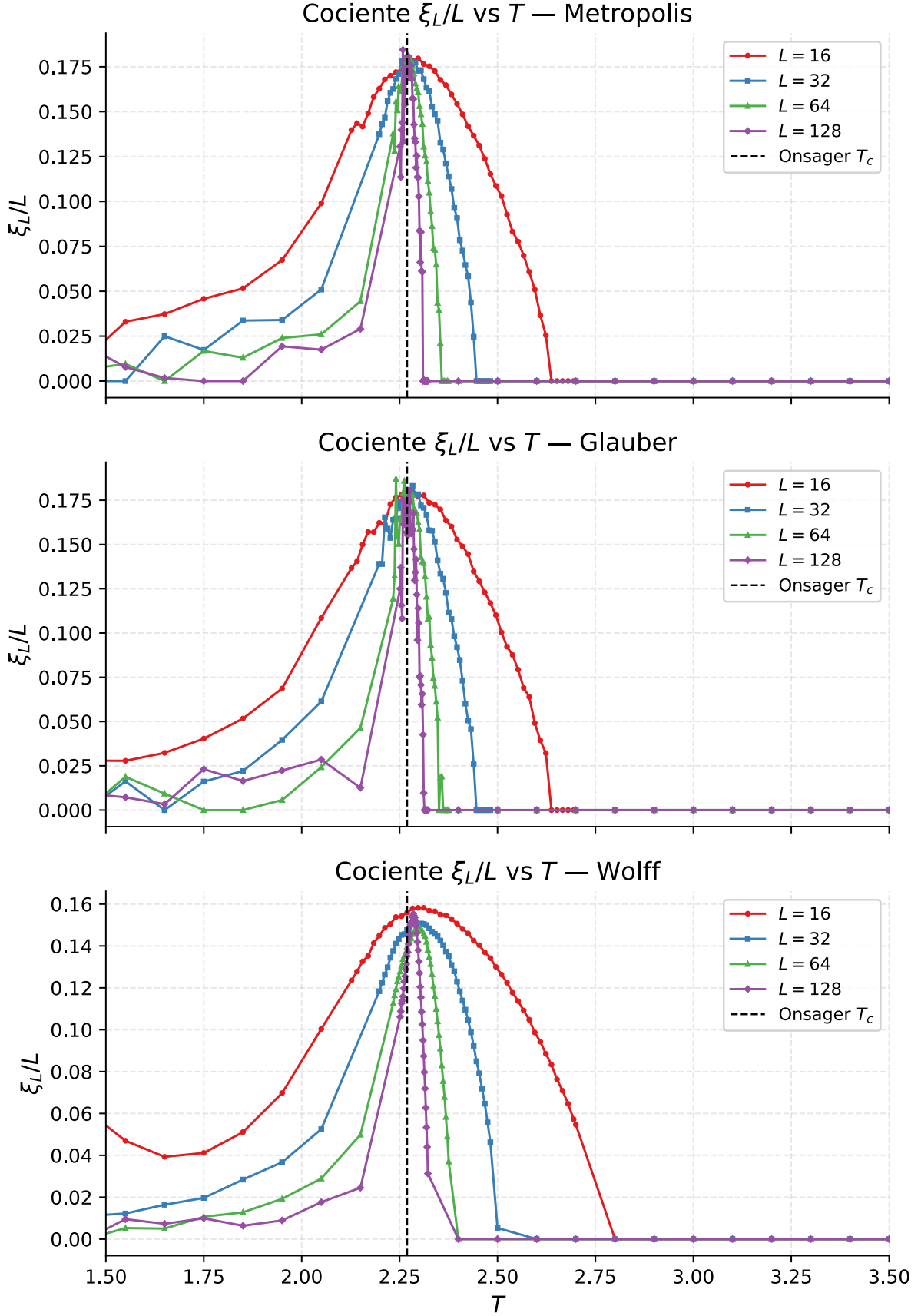


Figure 4.9: Scaled correlation length ξ_L/L vs T for each algorithm and $L \in \{16, 32, 64, 128\}$.

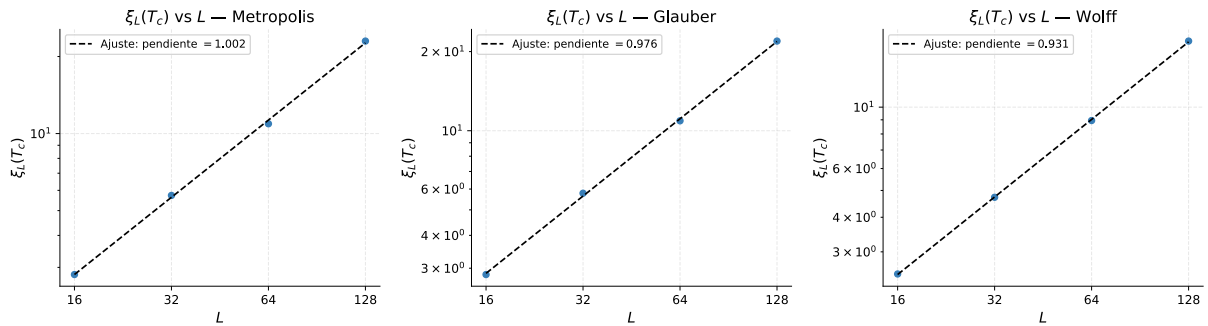


Figure 4.10: Linear fit of the correlation-length peak positions versus L in log–log scale.

4.6 Autocorrelation time

The plots in Fig. 4.12 are among the most informative for the algorithmic comparison.

There is a striking distinction. Metropolis and Glauber clearly exhibit critical slowing down: the peak near T_c grows with L and is orders of magnitude larger for $L = 128$ than for $L = 16$. Moreover, Glauber peaks are consistently about $3/2$ times larger than Metropolis, reflecting the generally lower acceptance probabilities in Glauber and hence longer autocorrelation times. By contrast Wolff displays very small τ_{int} (and larger statistical noise) across the range with only a tiny structure near T_c , directly illustrating that cluster algorithms essentially remove critical slowing down.

This behaviour explains the discrepancies between Wolff and the local algorithms in the data collapses. For truly decorrelated sampling at T_c Glauber would require taking measurements every ~ 9000 MCS rather than every 50 MCS used here, so the measurements near criticality for the largest sizes are strongly correlated and the statistical errors are underestimated.

By fitting the peak autocorrelation times to a power law $\tau_{\text{max}}(L) \sim L^z$ one can estimate the dynamic exponent z . With four sizes the trend for Metropolis and Glauber is clearly increasing and consistent with $z \approx 2$, although precise quantitative fits would require larger lattices and more samples to reduce statistical fluctuations. For Wolff estimating z is harder because τ_{int} is of order unity and noise dominates; the literature value $z \approx 0.25$ (Wolff, 1989) is difficult to distinguish from $z = 0$ with our data. Practically, obtaining an effectively independent sample near T_c costs orders of magnitude less with Wolff than with the local algorithms for the lattice sizes studied here. Figure 4.13 shows the fitted results, which are reasonably close to theoretical expectations.

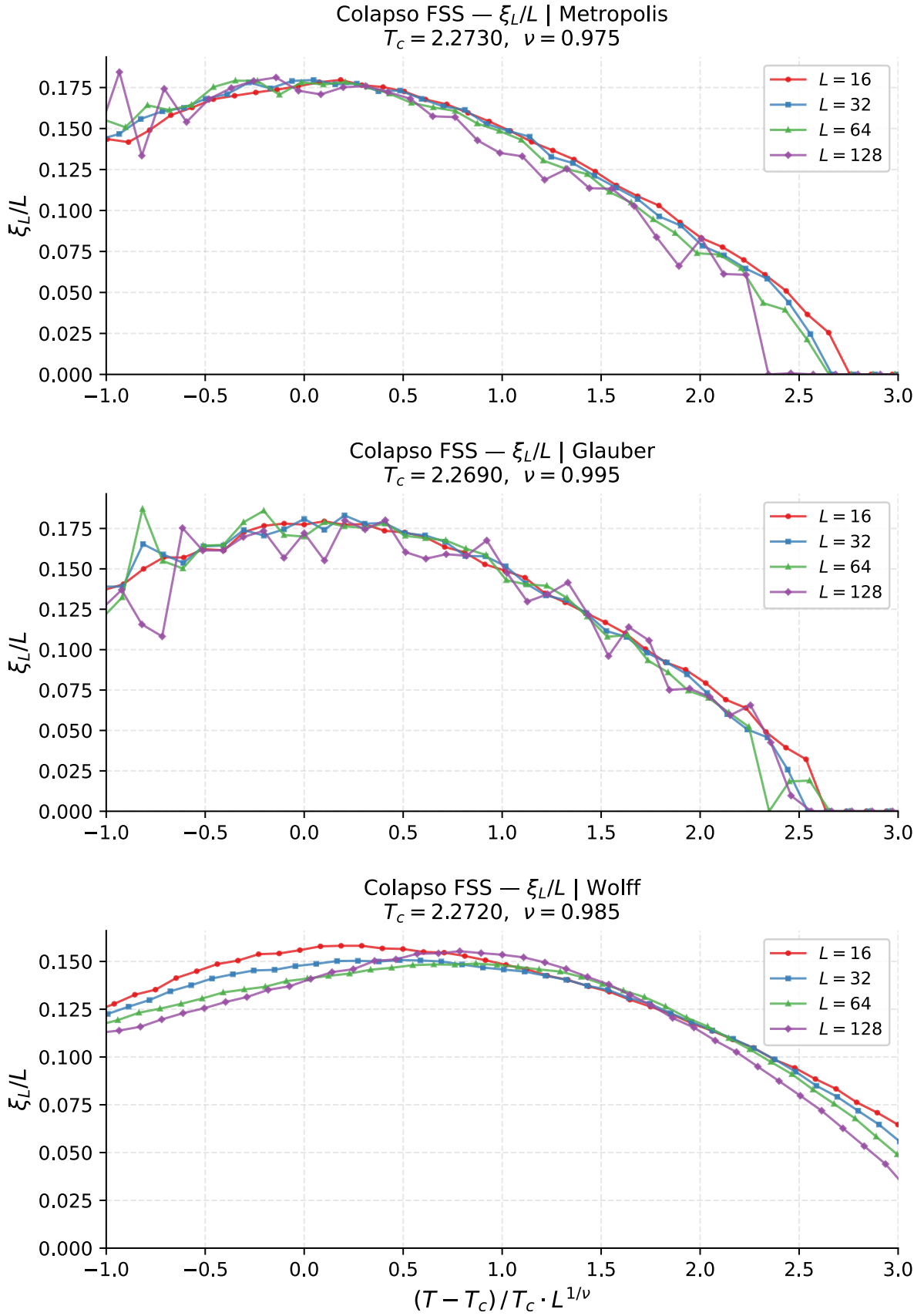


Figure 4.11: FSS collapse of ξ_L/L with experimental exponents for each algorithm.

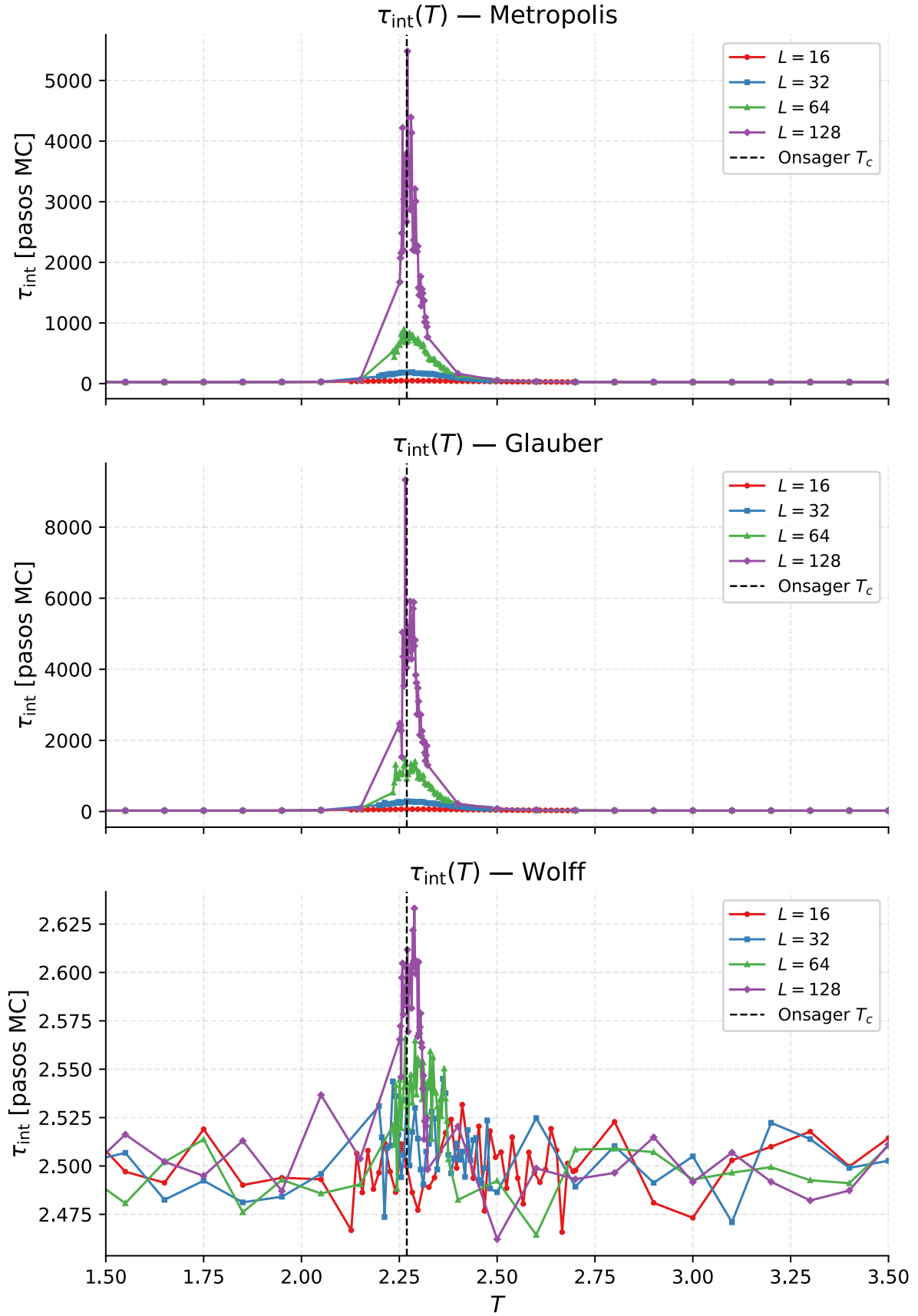


Figure 4.12: Integrated autocorrelation time τ_{int} vs T for each algorithm and $L \in \{16, 32, 64, 128\}$.

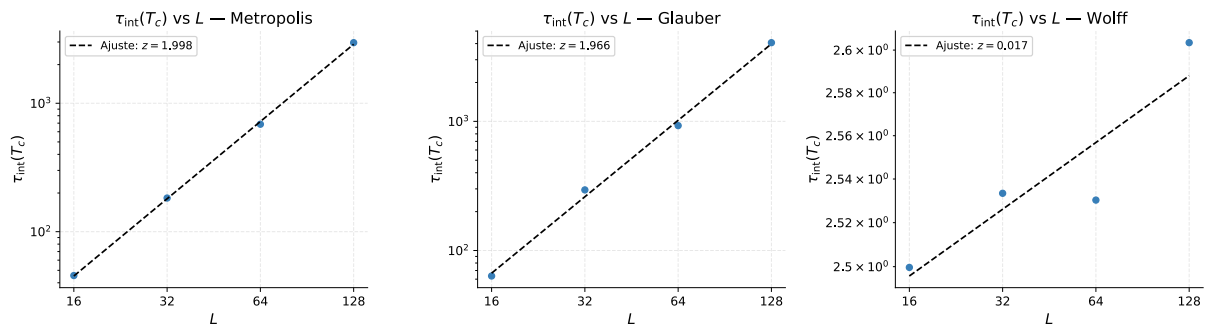


Figure 4.13: Logarithmic fit to estimate the dynamic exponent for each algorithm.

Chapter 5

Conclusions

In this work we have simulated the 2D Ising model on a square lattice with periodic boundary conditions, comparing three Monte Carlo algorithms (Metropolis, Glauber and Wolff) for $L \in \{16, 32, 64, 128\}$. Our aims were twofold: to verify that all three reproduce equilibrium physics and to compare their dynamical behaviour. In equilibrium the three algorithms agree, as required when sampling the same Boltzmann distribution. The Binder-cumulant crossings yield an estimate of T_c compatible with Onsager's result ($U_4^* \approx 0.61$), and the FSS collapse of χ and ξ_L/L using critical exponents confirms that the data belong to the universality class of the 2D Ising model. The main caveat arises near T_c for observables sensitive to autocorrelation: Metropolis and Glauber display critical slowing down, so a clean convergence of T_c with L is observed only for Wolff, and the peak position of the collapsed susceptibility shifts to $x \approx 1$ for local algorithms while Wolff's peak sits near $x \approx 1.5$, which we regard as the more reliable value.

The essential difference between the algorithms lies in dynamics as measured by the integrated autocorrelation time τ_{int} . Metropolis and Glauber suffer critical slowing down: the peak τ_{int} near T_c grows with L consistent with $z \approx 2$, and Glauber is somewhat slower than Metropolis (factor ~ 1.5 – 2) because Metropolis typically has higher acceptance rates and thus decorrelates faster. Wolff effectively removes critical slowing down: its τ_{int} is of order unity across the range (small z), so obtaining independent samples near T_c costs orders of magnitude less than with local updates — a decisive advantage for larger lattices.

As a natural extension, it would be interesting to repeat the comparison for the three-dimensional Ising model, where no exact solution exists and critical exponents are known only numerically, now that the simulation and data-analysis pipeline has been validated against the exact 2D results.

Conclusions (English)

In this work we have simulated the 2D Ising model on a square lattice with periodic boundaries, comparing three Monte Carlo algorithms (Metropolis, Glauber and Wolff) for $L \in \{16, 32, 64, 128\}$, with a twofold aim: to check that all three reproduce the equilibrium physics and to compare their dynamical behaviour. At equilibrium the three agree, as they must when sampling the same Boltzmann distribution. The estimate of T_c obtained from the Binder cumulant crossings is compatible with Onsager's value (with $U_4^* \approx 0,61$), and the FSS collapse of χ and ξ_L/L through the search for the critical exponents confirms that the data belong to the universality class of the two-dimensional Ising model. The only caveat appears near T_c , in the observables most sensitive to autocorrelation, where Metropolis and Glauber feel the critical slowing down, so that the clean convergence of T_c with L is seen only with Wolff and the peak of the collapsed susceptibility shifts to $x \approx 1$ for the local algorithms, against $x \approx 1,5$ for Wolff, which is the value we take as reliable.

The underlying difference between the algorithms lies in the dynamics, measured through the integrated autocorrelation time τ_{int} . Metropolis and Glauber suffer critical slowing down, with a peak of τ_{int} that grows with L around T_c consistently with $z \approx 2$; within this class, Glauber is somewhat slower than Metropolis (by a factor between 1,5 and 2), because Metropolis accepts the flips with a probability greater than or equal to that of Glauber for every ΔE and decorrelates sooner, which is a general result. Wolff, by contrast, practically eliminates the critical slowing down: its τ_{int} is of unit order over the whole range ($z \approx 0,017$), so that near T_c obtaining an independent sample costs several orders of magnitude less than with the local algorithms, an advantage that would become decisive for larger lattices.

As a natural continuation, it would be interesting to extend the comparison to the Ising model in three dimensions, where there is no exact solution and the exponents are known only numerically, now that we know that the program together with the data analysis works satisfactorily, as shown by comparing the results obtained against the exact results of the two-dimensional Ising model.

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